The derivative of a function

The definition, the notation, and what the number actually measures in geometry and in physics.

LESSON 5–6 | 2 × 40 MIN | ALGEBRA 11, §1.3 | P1 · 7.1–7.2

LESSON CLOCK

10' | 22' | 16' | 12' | 22' | 8'

The definition	10 min
First principles, three times	22 min
Two interpretations	16 min
Notation, and where it fails	12 min
Practice	22 min
Homework	8 min

LEARNING OBJECTIVES

- State the definition of the derivative as a limit of the difference quotient.
- Differentiate from first principles for a polynomial and a simple root.
- Interpret f′(x) as a gradient and as a rate of change.
- Use the three standard notations for the derivative.

TEXTBOOK

National	pp. 27–38
Cambridge	Pure Mathematics 1, pp. 122–133
Workbook	P1 Exercise 7A, 7B

01 Explanation

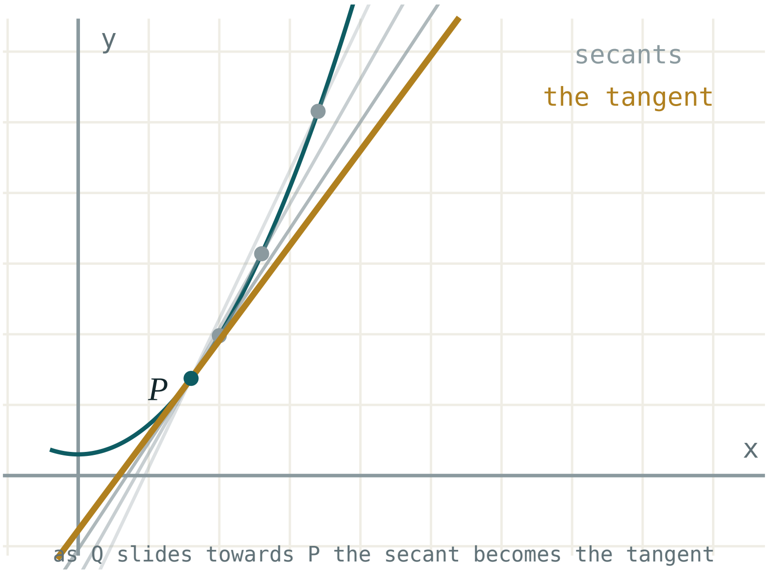
The definition

THE DERIVATIVE OF *F* AT *X*

$$f'(x) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}$$

when that limit exists. The function f is then **differentiable** at x.

Everything from the last two lessons is now in one line: the increments give the quotient, and the limit turns the secant into the tangent.



As h shrinks the secant becomes the tangent, and its gradient becomes f′(x).

Three derivatives from first principles

1. $f(x) = x^2$:

$$\frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \to 2x$$

2. $f(x) = x^3$:

$$\frac{(x+h)^3 - x^3}{h} = 3x^2 + 3xh + h^2 \to 3x^2$$

3. $f(x) = \frac{1}{x}$:

$$\frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \frac{-1}{x(x+h)} \to -\frac{1}{x^2}$$

THE PATTERN

$x^2 \to 2x$, $x^3 \to 3x^2$, $x^{-1} \to -x^{-2}$. In every case the index comes down in front and drops by one. That is the rule proved in the next lesson.

What the number means

CONTEXT	$F(X)$	$F'(X)$
geometry	height of the curve	gradient of the tangent
motion	displacement $s(t)$	velocity $v(t)$
motion, again	velocity $v(t)$	acceleration $a(t)$
economics	total cost $C(q)$	marginal cost
biology	population $P(t)$	growth rate

For the falling stone $s = 5t^2$: $s' = 10t$, so at $t = 2$ the speed is 20 m/s — the number the table in Lesson 1–2 was walking towards.

Notation, and where the derivative fails to exist

Three notations, all standard, all meaning the same thing:

$$f'(x) \cdot y' \cdot \frac{\mathrm{dy}}{\mathrm{dx}}$$

The Leibniz form $\frac{\mathrm{dy}}{\mathrm{dx}}$ keeps the increments visible and is preferred in physics and in the chain rule.

NOT EVERY FUNCTION HAS A DERIVATIVE EVERYWHERE

$f(x) = |x|$ has no derivative at 0: approaching from the left the quotient is -1, from the right $+1$. The two one-sided limits disagree, so no tangent exists — the graph has a **corner**. A vertical tangent, as in $y = \sqrt[3]{x}$ at 0, also fails.

02 Worked examples

EXAMPLE 1 **Differentiate $f(x) = 3x^2 - 5x$ from first principles.**

1. $f(x + h) = 3(x + h)^2 - 5(x + h)$
2. $f(x + h) - f(x) = 6xh + 3h^2 - 5h$
3. $\frac{\cdots}{h} = 6x + 3h - 5$
4. $h \to 0$

ANSWER

$f'(x) = 6x - 5$

EXAMPLE 2 **Differentiate $f(x) = \sqrt{x}$ from first principles.**

1. $\frac{\sqrt{x + h} - \sqrt{x}}{h}$

   Multiply by the conjugate.
2. $= \frac{h}{h(\sqrt{x + h} + \sqrt{x})}$
3. $= \frac{1}{\sqrt{x + h} + \sqrt{x}}$
4. $\to \frac{1}{2\sqrt{x}}$

ANSWER

$f'(x) = \frac{1}{2\sqrt{x}}$

EXAMPLE 3 **A body moves with $s = t^3 - 6t^2$ metres. Find its velocity and acceleration at $t = 4$.**

1. $v = s' = 3t^2 - 12t$

2. $v(4) = 48 - 48 = 0$

   Momentarily at rest.

3. $a = v' = 6t - 12$

4. $a(4) = 12$

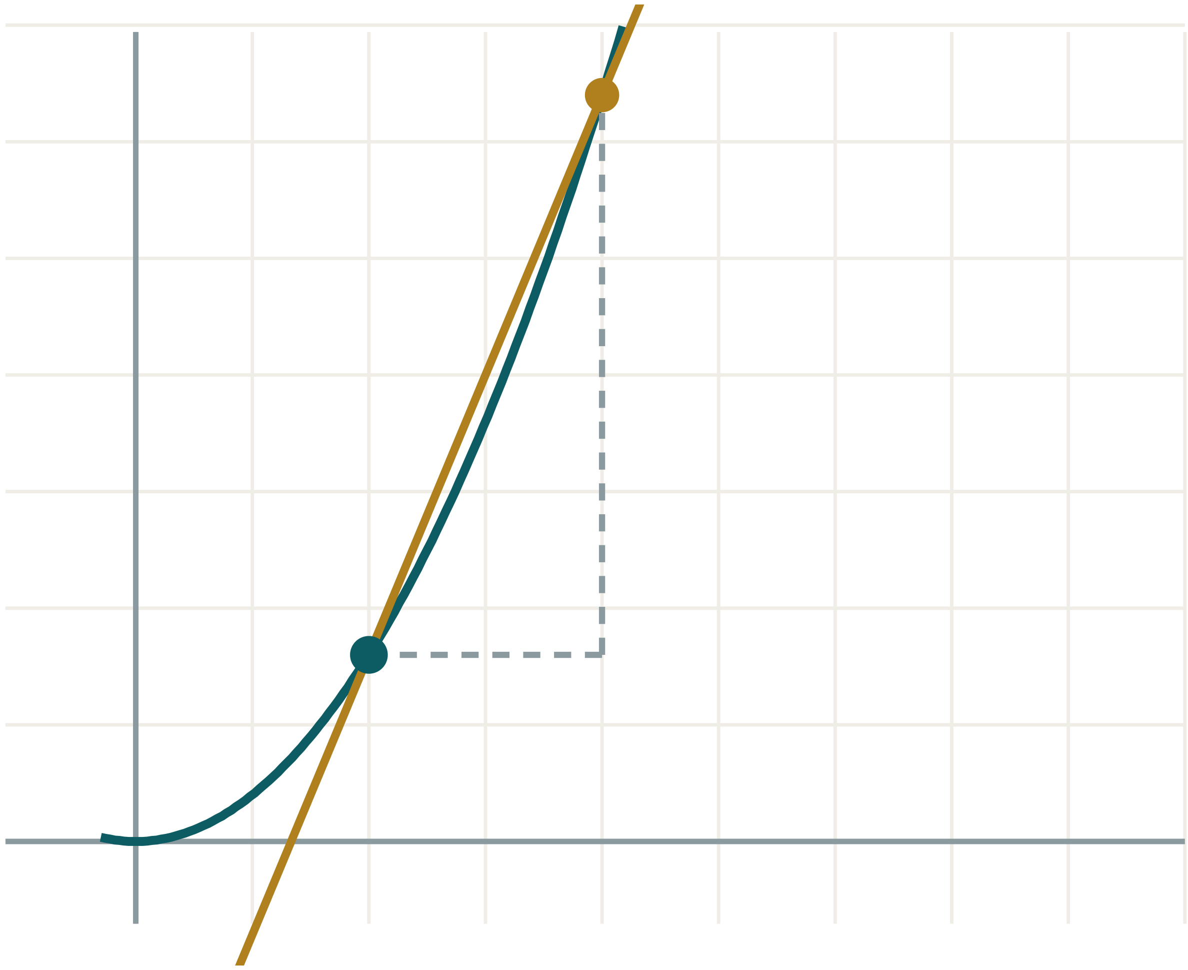
ANSWER

$v = 0$ m/s, $a = 12$ m/s²

03 Interactive model

Set h to 1, then 0.1, then 0.01 and read the gradient converging.

The derivative as a limit



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Derivative	Hosila	Производная
Differentiation	Differensiallash	Дифференцирование
From first principles	Ta’rif bo‘yicha	По определению
Gradient function	Burchak koeffitsienti funksiyasi	Функция углового коэффициента
Differentiable	Differensiallanuvchi	Дифференцируемая
Instantaneous velocity	Oniy tezlik	Мгновенная скорость
Leibniz notation	Leybnits belgilanishi	Обозначение Лейбница
Corner point	Sinish nuqtasi	Угловая точка
Rate of change	O‘zgarish tezligi	Скорость изменения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The derivative is the limit of:

- the function
- the secant gradient
- the tangent
- the increment

From first principles, $(x^3)'$ is:

- $3x$
- $3x^2$
- x^2
- $3x^3$

$f'(x)$ for $f(x) = \frac{1}{x}$ is:

- $\frac{1}{x^2}$
- $-\frac{1}{x^2}$
- $\ln x$
- $-x$

$|x|$ at $x = 0$:

- has derivative 0
- has derivative 1
- has no derivative
- has derivative −1

If $s(t)$ is displacement, $s'(t)$ is:

- distance
- velocity
- acceleration
- time

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. From first principles, $(x^2)'$

   ANSWER $2x$

2. From first principles, $(5x)'$

   ANSWER 5

3. From first principles, $(7)'$

   ANSWER 0

4. $f(x) = x^2$; find $f'(3)$

   ANSWER 6

5. $f(x) = x^3$; find $f'(2)$

   ANSWER 12

6. $s = 5t^2$; find $v(2)$

   ANSWER 20 m/s

7. Name three notations for the derivative

   ANSWER $f'(x), y', \frac{\mathrm{dy}}{\mathrm{dx}}$

Medium · the standard question

7 PROBLEMS

1. First principles: $(2x^2 + x)'$

   ANSWER $4x + 1$

2. First principles: $(x^2 - 3x + 1)'$

   ANSWER $2x - 3$

3. First principles: $\left(\frac{1}{x}\right)'$

   ANSWER $-\frac{1}{x^2}$

4. First principles: $(\sqrt{x})'$

   ANSWER $\frac{1}{2\sqrt{x}}$

5. $s = t^3 - 6t^2$; find $v(4)$

   ANSWER 0

6. Same; find $a(4)$

   ANSWER 12

7. Where does $|x - 2|$ fail to be differentiable?

   ANSWER $x = 2$

Hard · stretch and reason

7 PROBLEMS

1. First principles: $(x^4)'$

ANSWER $4x^3$

2. First principles: $\left(\frac{1}{x^2}\right)'$

ANSWER $-\frac{2}{x^3}$

3. First principles: $\left(\frac{1}{\sqrt{x}}\right)'$

ANSWER $-\frac{1}{2x\sqrt{x}}$

4. Show $f(x) = |x|$ has one-sided derivatives ± 1 at 0

ANSWER Quotient is $\frac{|h|}{h}$

5. $s = t^3 - 3t^2 + 2t$; when is the body at rest?

ANSWER $t = 1 \pm \frac{1}{\sqrt{3}}$

6. A differentiable function is always continuous. Give a continuous function that is not differentiable

ANSWER $|x|$ at 0

7. From first principles find $(x^{-3})'$

ANSWER $-3x^{-4}$

07 Homework

Homework — 5 tasks

Tasks 1–3 must be done from the definition, not from a rule.

1. From first principles, differentiate $f(x) = 4x^2 - 7x + 2$.

2. From first principles, differentiate $f(x) = \frac{2}{x}$.
3. From first principles, differentiate $f(x) = \sqrt{2x}$.
4. $s = 2t^3 - 9t^2 + 12t$. Find the velocity and the times at which the body is at rest.
5. Explain, with a sketch, why $y = |x + 1|$ has no derivative at $x = -1$.